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Efficient upgrading of polyolefin plastics into C₅–C₁₂ gasoline alkanes over a Pt/W/Beta catalyst†

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A Pt/W/Beta catalyst was developed for direct and efficient hydrogenolysis of low density polyethylene (LDPE) into C₅–C₁₂ gasoline alkanes with a high yield of 63.6 wt% (yield of C₁–C₁₃ light alkanes is 94.0 wt%) at 250 °C and 3.0 MPa H₂ within 1.0 h. The obtained gasoline alkanes have a narrow carbon number distribution and contain abundant mono-branched isomers, which were attributed to well-dispersed Pt species and proximity of the cooperating active sites, especially the moderate acidity upon the introduction of 0.5 wt% tungsten species. The catalyst was also applicable to hydrogenolysis of light low density polyethylene (LLDPE), high density polyethylene (HDPE) and polypropylene (PP) to obtain the valuable gasoline alkanes (C₅–C₁₂) in high yield (50.8–73.5 wt%). The C–C bond cleavage of plastics over the Pt/W/Beta catalyst was found to occur through medium to long-chain hydrocarbons (*n*-C_{*x*}, *x* > 13) based on experiments performed using docosane (*n*-C₂₂ alkane) as the model feedstock.

Plastics are widely used in various fields due to their inherent advantages, but accumulated and refractory plastic waste especially microplastics causes serious global white pollution and threatens human health.^{1–3} To mitigate this situation, thermochemical decomposition (ranging from 400 to 900 °C) was exploited to convert plastics of high calorific value into monomers and added-value gases, fuels and waxes, which can be used as the feedstock for producing new plastics, leading to energy recycling and reduction of environmental risks.^{4–7}

Recently, hydrogenolysis has been found to be a viable solution for activating the C–C bonds on the polymer backbone over heterogeneous catalysts, leading to the high yield formation of liquid phase products under mild conditions.^{6,8–10} Various noble-metal catalysts with additives and acid supports were reported to be effective for the direct hydrogenolysis of plastics.^{11–21} Liquid fuels (C₅–C₂₁) with 77.0 wt% yield and a wide carbon number distribution were obtained over the Ru/CeO₂ catalyst at 200 °C within 8.0 h with low density polyethylene (LDPE) as the feedstock.²² To obtain a narrow carbon number distribution for the products, Tennakoon *et al.* developed a mesoporous mSiO₂/Pt/SiO₂ catalyst and 79.0 wt% yield of liquid products ranging from C₉ to C₁₈ was obtained using high density polyethylene (HDPE) as the feedstock in 8.0 h at 250 °C.²³ Yao *et al.* reported the conversion of HDPE to C₁–C₃ hydrocarbons with >95.0% selectivity using plasma-activated H₂ over combined Pt/C and SAPO-34 zeolite catalysts at 300 °C.²⁴ Liu *et al.* further demonstrated that acidity plays an important role in the catalytic activity and product selectivity for the hydrogenolysis reaction, and they used mixed Pt/WO₃/ZrO₂ and FAU-type zeolites for the conversion of LDPE and obtained gasoline alkanes (C₅–C₁₂) with a yield of 72.0 wt% at 250 °C and 3.0 MPa H₂ in 2.0 h.¹⁷ Thus, it is highly desirable to develop a single efficient bifunctional heterogeneous catalyst with moderate acidity for the conversion of LDPE into liquid alkanes with a narrower carbon number distribution and a short reaction time.

In this work, we developed a bifunctional Pt/W/Beta catalyst to convert LDPE into gasoline alkanes (C₅–C₁₂) with a yield of 63.6 wt%, where the total yield of alkanes (C₁–C₁₃) reached 94.0 wt% at 250 °C and 3.0 MPa H₂ in 1.0 h. Furthermore, the role of Brønsted acid sites derived from the W species and Beta support was revealed by catalyst characterization and performance analysis. The C–C bond cleavage mechanistic pathway of LDPE was proposed based on experiments using C₂₂ alkane as the model feedstock, where the role of Pt, W and Beta was revealed. The applicability of the Pt/W/Beta catalyst for the conversion of linear low-density polyethylene (LLDPE), HDPE

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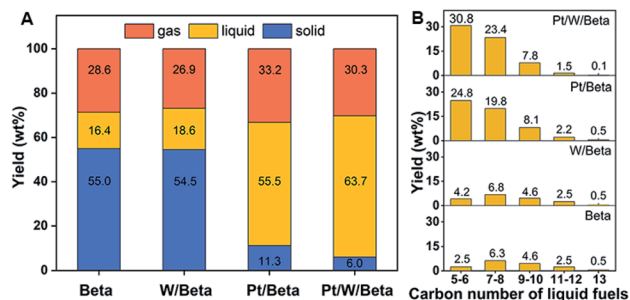


Fig. 1 Hydrogenolysis of LDPE (A) and carbon number distribution of liquid fuels (B) over Beta, W/Beta, Pt/Beta and Pt/W/Beta catalysts. (Reaction conditions: catalyst (Pt: 2.0 wt%, W: 0.5 wt%) 50 mg, LDPE 2.0 g, H_2 3.0 MPa, 250 °C, 1.0 h).

and polypropylene (PP) was also displayed with a high yield of gasoline alkanes through the hydrogenolysis reaction.

Fig. S1† shows the influence of the amount of catalyst, reaction time, temperature and H_2 pressure on the performance of our Pt/W/Beta catalyst for LDPE hydrogenolysis. At 250 °C and 3.0 MPa H_2 in 1.0 h, the yield of gasoline alkanes (C_5 – C_{12}) reached 63.6 wt%, whereas that of total alkanes (C_1 – C_{13}) reached 94.0 wt% (Fig. 1A, S2, S3 and S4A†). The carbon number of the liquid fuels was in the range of 5–13 during hydrogenolysis of the Pt/W/Beta catalyst, compared with 10–25 during direct pyrolysis (Fig. S5†). When the Beta zeolite and W/Beta served as catalysts, solid residues were the dominant products with a yield of more than 50.0 wt%, while the liquid hydrocarbons account for less than 20.0 wt% (Fig. S3 and S4B†). The amount of C_5 – C_{12} hydrocarbons increased from 15.9 wt% to

55.0 wt% when Pt/Beta was employed as the catalyst compared with the Beta zeolite, as Pt hydrogenated the intermediates and suppressed the formation of char residues due to the facilitation of the C–C bond breaking under a H_2 atmosphere.¹⁷ Compared to the Pt/Beta catalyst, more gasoline alkanes (63.6 wt%) along with 54.2 wt% C_5 – C_8 alkanes were obtained over the Pt/W/Beta catalyst (Table S1†), suggesting that the W additive favoured the formation of gasoline alkanes. Meanwhile, we found that the amount of specific linear chain molecules and their mono-branched isomers (2-methyl-butane, *n*-pentane, 2-methyl-pentane, 3-methyl-pentane, *n*-hexane, 2-methyl-hexane, 3-methyl-hexane, *n*-heptane, 2-methyl-heptane, 3-methyl-heptane, and *n*-octane) in the liquid products increased significantly, and the mono-branched isomers of C_5 – C_8 alkanes made up 61.1%, which were ideal components of high-grade gasoline (Table S2†).

The Pt/W/Beta catalyst exhibited catalytic performance with the yield of the C_5 – C_{12} gasoline alkanes reaching 63.6 wt% and that of solid residues limited to 6.0 wt% due to its structure. Catalyst characterization studies showed that Pt and W metals were dispersed evenly on the multilayer Beta support (Fig. 2A and B) according to the similar N_2 adsorption and desorption plots of both Beta and Pt/W/Beta catalysts (Fig. 2C), as well as the disappearance of the XRD peaks of Pt and W species on the Pt/W/Beta catalyst (Fig. 2D). Meanwhile, the size of Pt particles was estimated to be 1.2 nm on average from Fig. 2B. From the H_2 -TPD and H_2 -TPR plots shown in Fig. 2E and F, we found that the Pt species over the Pt/W/Beta and Pt/Beta catalysts were both reduced below 200 °C and showed excellent H_2 adsorption and dissociation ability reflected by the peaks at 120 °C and 430 °C, which were assigned to the desorption of hydrogen from

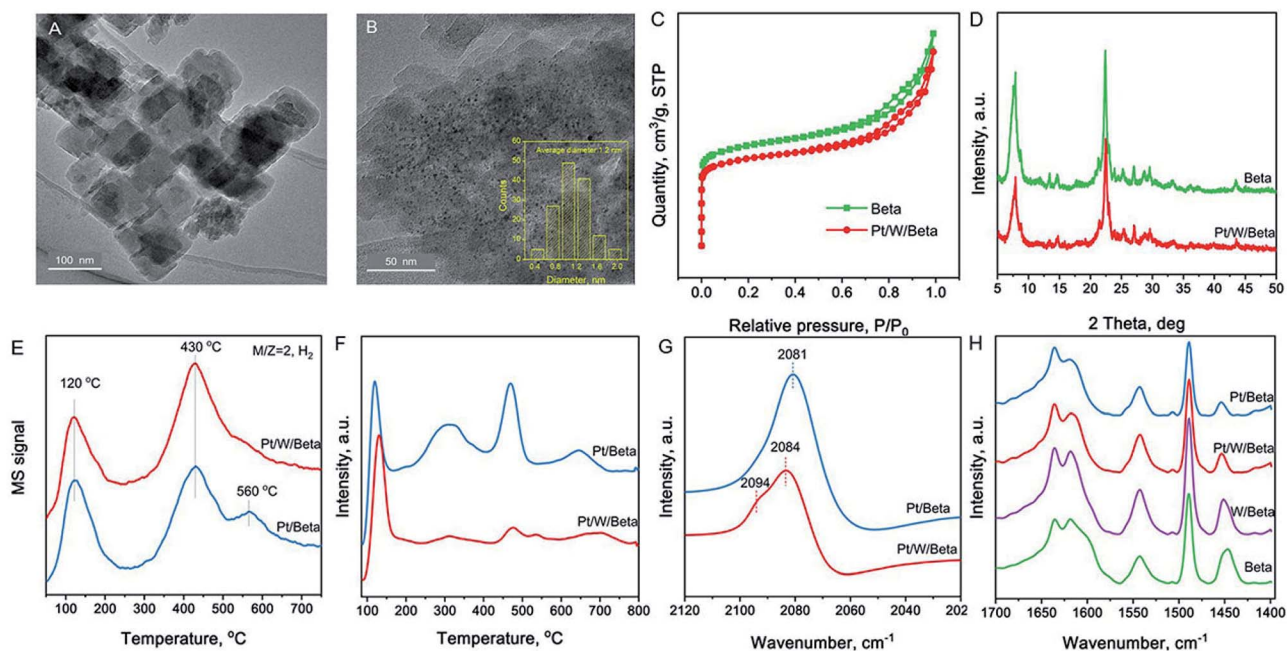


Fig. 2 Characterization of the synthesized catalyst. (A) and (B) TEM and HRTEM images of the Pt/W/Beta catalyst; (C) N_2 adsorption and desorption plots and (D) XRD patterns of Beta zeolites and the Pt/W/Beta catalyst; (E) H_2 -TPD and (F) H_2 -TPR spectra of the Pt/Beta and Pt/W/Beta catalysts; (G) CO drift of the Pt/Beta and Pt/W/Beta catalysts; (H) Py-IR spectrum of the synthesized catalyst.

metallic Pt surfaces and the spillover hydrogen. These hydrogen species favor the cracking of PE into medium-long-chain molecules and their further conversion into light alkanes when compared with the Beta zeolite and Pt/Beta catalysts (Fig. 1). Because the addition of W promoted the reduction and the dispersion of Pt species over the catalysts based on Fig. 2F, G and S6,[†] at the same time brought more Brønsted acid sites into the catalysts according to Fig. 2H, the Pt/W/Beta catalyst showed a higher yield of gasoline alkanes with less solid residues. The stability of the Pt/W/Beta catalyst was investigated at 250 °C and 3.0 MPa H₂ in 1.0 h with the regeneration of the recovered catalyst between each experiment. After five successive runs, no significant decline in LDPE conversion and alkane yields were observed, indicating the stability of the Pt/W/Beta catalyst (Fig. S7[†]). Meanwhile, the XRD patterns of the spent Pt/W/Beta catalyst display peaks of the typical Beta phase with a BEA-type structure, indicating that the framework structure of the Beta zeolite after the reaction did not change (Fig. S8[†]).

Characterization of the Pt/W/Beta catalyst showed that the addition of the W species leads to the change in the amount of Brønsted acid sites (Fig. 3A and S9[†]). Brønsted acid sites derived from the additive and the support have obvious influence on the hydrogenolysis of LDPE, which initiated the cleavage of the C–C bond by protonation of the feedstocks.^{6,17,25,26} Compared with the Pt/Beta catalyst (Fig. 3B and Table S3[†]), the addition of 0.2 wt% W increased the yield of C₅–C₁₂ gasoline alkanes from 55.0 wt% to 61.8 wt%, and this value reached 63.6 wt% when 0.5 wt% W species was introduced. In the meantime, the mono-branched isomers of C₅–C₈ alkanes made up 61.1%, and the introduced Brønsted acid led to the increase of the isomers by 15.0% compared to the Pt/Beta catalyst (Tables S3 and S4[†]). When the W content reached 1.0 wt%, the catalyst with high acidity produced more gas alkanes at the expense of liquid alkanes with the yield of gasoline alkanes reduced to 58.9 wt%, consistent with the previous studies.^{25,27} Meanwhile, as shown in Fig. S10 and S11,[†] the Beta zeolite displayed better support performance for improving the yield of gasoline alkanes compared with the USY zeolite, LaY zeolite and Al₂O₃ in LDPE hydrogenolysis, which was attributed to the suitable acidity (derived from the Beta support and W species) and the latter was in favor of C–C bond cleavage in LDPE into C₅–C₁₂ gasoline alkanes (Tables S5 and S6[†]). When the Pt/W/Al₂O₃ catalyst was

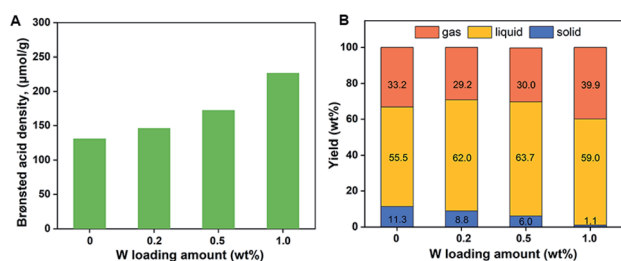


Fig. 3 Brønsted acid site density (desorption temperature of 350 °C) of the W/Beta catalysts with different W loadings (A) and product yields during hydrogenolysis of LDPE over the catalysts with different W loadings (B). (Reaction conditions: Pt/W/Beta (Pt: 2.0 wt%) 50 mg, LDPE 2.0 g, H₂ 3.0 MPa, 250 °C, 1.0 h).

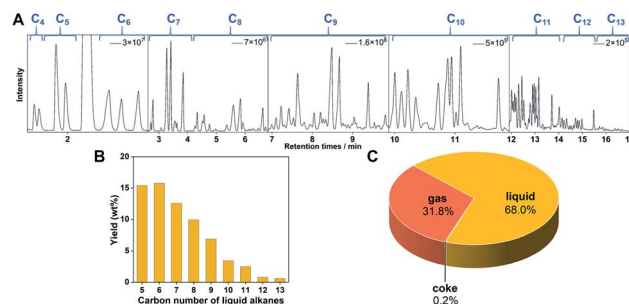


Fig. 4 Hydrogenolysis of C₂₂H₄₆ over the Pt/W/Beta catalyst. GC-MS results (A) and yield of liquid alkanes for different carbon numbers (B), and distribution of all products (C). (Reaction conditions: catalyst (Pt: 2.0 wt%, W: 0.5 wt%) 50 mg, docosane 2.0 g, H₂ 3.0 MPa, 250 °C, 1.0 h).

blended with the Beta zeolite, the yield of the gasoline alkanes (C₅–C₁₂) was 42.2 wt%. The higher efficiency of the Pt/W/Beta catalyst was attributed to the close proximity of the Pt components to the acid sites on the Beta zeolite support (Fig. S12[†]).

Docosane (*n*-C₂₂ alkane) was further used as the model feedstock to reveal the potential mechanistic pathway for the cleavage of the C–C bond over the Pt/W/Beta catalyst. As shown in Fig. 4, the reaction of the *n*-C₂₂ alkane at 250 °C and 3.0 MPa H₂ in 1.0 h had similar performance to LDPE hydrogenolysis. The products included 31.8 wt% C₁–C₄ gaseous alkanes as well as 67.3 wt% C₅–C₁₃ liquid alkanes (Fig. 4A), where the main liquid C₅–C₈ alkanes accounted for 54.0 wt% (Fig. 4B). The similarity in the product distribution to LDPE hydrogenolysis indicated that medium-long-chain hydrocarbons (*n*-C_{*x*}, *x* > 13) were potential intermediates, which were further converted into a series of light alkanes (C₁–C₁₃) over the Pt/W/Beta catalyst.

Based on the hydrogenolysis behaviour of docosane over the Pt/W/Beta catalyst, the possible pathway for the hydrogenolysis of LDPE on the Pt/W/Beta catalyst is depicted in Fig. 5. LDPE was broken into medium-long-chain *n*-alkanes and *n*-olefins (*n*-C_{*x*}, *x* > 13) at the metal active sites,¹⁷ which was attributed to the excellent H₂ adsorption and dissociation ability of Pt species on the Pt/W/Beta catalyst (Fig. 2). The *n*-olefins desorbed from the metal active sites, and were further protonated to form the alkylcarbenium ions (*n*-C_{*x*}⁺) after diffusing to the Brønsted acid sites.²⁸ The *n*-alkylcarbenium ions were then transformed into mono-branched alkylcarbenium ions (iso-C_{*x*}⁺) *via* isomerization. Some of the mono-branched olefins further diffused to the metal active sites, and were hydrogenated to mono-branched

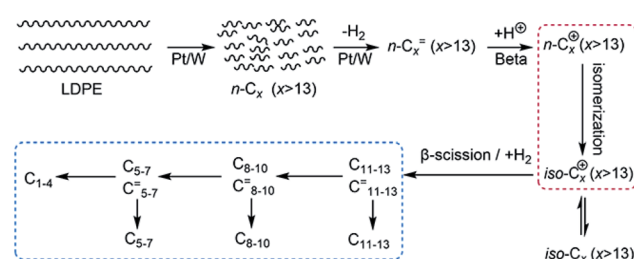


Fig. 5 Proposed hydrogenolysis pathway of LDPE over the Pt/W/Beta catalyst.

Table 1 Yields and carbon number distribution of liquid products from LLDPE, HDPE and PP hydrogenolysis^a

Plastics	Yield/wt%			Yield of liquid alkanes/wt%				
	Gas	Liquid	Solid	C ₅ –C ₆	C ₇ –C ₈	C ₉ –C ₁₀	C ₁₁ –C ₁₂	C ₁₃
LLDPE	31.8	65.0	3.2	34.4	24.3	5.8	0.5	<0.1
HDPE	25.2	74.5	0.3	31.0	25.3	12.4	4.9	1.0
PP	47.1	52.3	0.6	19.9	16.6	9.5	4.8	1.5

^a (Reaction conditions: Pt/W/Beta (Pt: 2.0 wt%, W: 0.5 wt%) 50 mg, plastic 2.0 g, H₂ 3.0 MPa, 250 °C, 1.0 h).

alkanes (iso-C_x). *n*-C_x⁺ and iso-C_x⁺ underwent β-scission, deprotonation, and hydrogenation reactions, leading to C_{11–13} and C_{5–7} alkanes or olefins, which would repeat the above steps to produce CH₄, C_{2–4}, C_{5–7} and C_{8–10} alkanes or olefins. The appropriate proximity of the Pt species and the Brønsted acid sites favors this process.²⁹ Under the H₂ atmosphere favoring complete hydrogenation, the C_{1–13} alkanes were eventually formed over the Pt/W/Beta catalyst.

To evaluate the applicability of the Pt/W/Beta catalyst, hydrogenolysis of LLDPE, HDPE and PP was carried out at 250 °C and 3.0 MPa H₂ pressure in 1.0 h (Table 1). The hydrogenolysis of LLDPE showed similar reactivity to that of LDPE, resulting in 31.8 wt% gaseous alkanes and 65.0 wt% liquid alkanes. When HDPE served as the feedstock, the yield of C₅–C₁₂ gasoline alkanes reached 73.6 wt% over the Pt/W/Beta catalyst, among which C₅–C₁₀ alkanes were the dominant products and accounted for 68.7 wt%. PP hydrogenolysis also resulted in gasoline alkanes with the yield of 50.8 wt% along with 47.1 wt% of gaseous alkanes. For comparison, the hydrogenolysis results of polyolefin plastics in different catalytic processes are listed in Table S7.† It showed that the Pt/W/Beta catalyst has significant advantages in terms of reaction time and carbon number distribution of products. The above results indicated that the Pt/W/Beta catalyst exhibited performance for converting various plastic feedstocks into gasoline alkanes with a narrow carbon number distribution under mild conditions.

Conclusions

In summary, efficient and direct hydrogenolysis of LDPE into C₅–C₁₂ gasoline alkanes over the Pt/W/Beta catalyst was exploited, which gave a yield of 63.6 wt% for C₅–C₁₂ gasoline alkanes along with 30.3 wt% of C₁–C₄ gaseous alkanes at 250 °C and 3.0 MPa H₂ in 1.0 h. Nearly 94.0 wt% of the feedstock was converted into alkanes, which is attributed to the high dispersion of Pt species, the moderate acidity, the proximity of the cooperating active sites, and the stable structure of the Pt/W/Beta catalyst. The medium-long-chain hydrocarbons (*n*-C_x, *x* > 13) were proved as potential intermediates, which were further converted into a series of C₁–C₁₃ light alkanes over the Pt/W/Beta catalyst when docosane (*n*-22 alkane) served as the model feedstock. Meanwhile, LLDPE, HDPE and PP can also be effectively converted into C₅–C₁₂ gasoline alkanes over the Pt/W/Beta

catalyst with yields of 65.0 wt%, 73.6 wt% and 50.8 wt%, respectively. Thus, the Pt/W/Beta catalyst showed performance for converting various plastic wastes into gasoline alkanes with a narrow carbon number distribution and abundant mono-branched isomers, which helps to increase the added value of the recycled plastics.

Conflicts of interest

There are no conflicts to declare.

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